CTU training solutions | Private Tertiary Education

CAPSTONE PROJECT

SOFTWARE TEST PLAN

Student / Group Template

Project Title	SmartDesk: A Customer Support Ticketing System for Small IT Support Teams
Group Members	Freek Schaefer-Gerber, Wian Sandenbergh
Module	SPM622 / MDB622
Lecturer / Supervisor	Mr. Dunstan
Week / Stage	Testing & Evaluation
Submission Date	29 October 2026

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1. Test Objectives

Define what testing must demonstrate.

To verify that SmartDesk meets all Must-have functional requirements (authentication, customer isolation, ticket CRUD, assignment, and comments) and non-functional requirements (security, performance, usability). To ensure the system correctly enforces role-based permissions and to ensure no Critical or High defects remain open before final submission.

2. Scope

State features included and excluded.

- **Included:** User registration and login, role-based access control (Customer, Agent, Admin), customer data isolation, ticket creation and validation, staff-only ticket assignment, status and priority management, attributable comments, and the status dashboard. Security testing (CSRF, SQL injection).
- **Excluded:** Email notifications, file attachments, SLA timers, and audit history tables, as these are defined as out of scope for the MVP.

3. Test Environment

Document hardware, software, browsers/devices, database and test data.

- Hardware/OS: Local Windows development machine.
- Software Stack: Python 3.11+, Flask 3, SQLite 3.
- Testing Tools: pytest (automated), Chrome/Edge developer tools (manual).
- Test Data: Synthetic database generated via `flask init-db` and `flask seed-demo`. The automated suite uses isolated temporary databases (`tmp_path`) per test to ensure repeatability.

4. Test Types

Identify unit, integration, system, usability, security or performance tests as applicable.

- Unit / Integration: Automated pytest suite testing authentication, authorisation, and ticket creation.
- System: Manual, scripted walkthroughs of the full user journey across all three roles.
- Security: Manual, adversarial testing (CSRF omission, cross-customer access, unauthorised role action, injection inputs).

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- Usability: Observation of first-time users attempting to create a ticket without instruction.

5. Entry and Exit Criteria

Define when testing begins and what constitutes completion.

- Entry Criteria: Application starts successfully, database schema initialises, and demo accounts are seeded.
- Exit Criteria: All automated tests pass, all Must-have requirements are exercised manually, and no Critical or High defects remain open.

6. Roles and Responsibilities

Assign testing responsibilities.

- Wian Sandenbergh: Automated testing (pytest), security testing, and defect logging.
- Freek Schaefer-Gerber: Usability observation, system walkthroughs, and validation message review.
- Both: Manual scripted system journeys across all roles.

7. Defect Management

Explain how defects are recorded, prioritised, fixed and retested.

Defects are recorded in the `Defect_Log.xlsx` file. They are prioritised by severity (Critical, High, Medium, Low) based on impact to the system. Assigned team members fix the defects, commit the changes to Git, and retest the specific issue to confirm resolution before closing the ticket.

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